Final Report

Forecasting Crime Rates in the City of Fort Worth Using Time Series Models

University of Texas at Arlington

Dr. Christopher Candreva

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Ankita Malu

Kaushal Dhungel

1. Introduction

Crime prediction is a central topic in urban economics and public policy because accurate forecasts allow police departments to allocate resources, schedule patrols, and plan preventive interventions more effectively. Understanding how crime changes over time is particularly important for large metropolitan areas, where shifts in population, economic conditions, and seasonal patterns create substantial variation in criminal activity. In cities such as Fort Worth, crime displays clear and recurring seasonal cycles, with higher activity during warmer months and lower levels in winter. These characteristics make crime data well-suited for time series analysis. This study aims to forecast monthly crime levels in Fort Worth, Texas, and to provide insight into expected trends over the next eighteen months.

To conduct this analysis, daily incident-level crime data were aggregated into monthly totals and examined for long-term trends, seasonal behavior, and potential non-stationarity. Preliminary steps included visual inspection of the raw series, decomposition to separate trend and seasonal components, and statistical testing to determine whether differencing or variance-stabilizing transformations were necessary. Based on the exploratory analysis, several forecasting models were estimated, including exponential smoothing (ETS) models and a range of SARIMA specifications. In addition, a dynamic regression model incorporating temperature as an exogenous variable was evaluated to test whether weather-related factors improved forecast performance. Model selection was guided by information criteria, accuracy on the test sample, and diagnostic checks to confirm that the residuals behaved like white noise.

The final forecasts were produced using the model that best captured the structure of crime in Fort Worth, which was a SARIMA (1,1,1)(1,1,1) [12] model incorporating both annual seasonality and short-run crime dynamics. This model generated monthly crime predictions for

the following eighteen months, along with their corresponding prediction intervals. The forecasts suggest that historical seasonal patterns are likely to continue, with expected increases during summer and declines in winter. These results provide a useful reference point for anticipating how crime may evolve in the near future and offer a data-driven foundation for planning and policy discussions.

2. Background

Research on crime and its economic and social determinants dates back several decades. In one of the earliest influential contributions, Isaac Ehrlich developed a formal economic model in which individuals decide whether to commit crimes by balancing potential benefits against the costs associated with detection and punishment (Ehrlich 1973). This model helped establish crime as a phenomenon that can be studied and forecasted using quantitative and theoretical tools rather than only sociological or qualitative analysis.

More recently, academic research has turned to time series forecasting of crime. Some studies rely on univariate models. For instance, using monthly police records from London, Islam & Raza (2020) found that ARIMA models can produce reasonably accurate two-year forecasts of aggregate crime and often outperform simpler benchmarks based on exponential smoothing. Other work emphasizes the role of seasonality. In particular, Delgado et al. (2023) show that seasonal ARIMA models improve forecast accuracy substantially when data exhibit strong recurring patterns.

A second strand of research examines how weather, and temperature in particular, shapes crime dynamics. Using daily data for Dallas, Texas, Gamble and Hess (2012) show a non-linear but generally positive association between ambient temperature and violent crime (Gamble & Hess, 2012). More recent time-series analyses in South Korea and other settings confirm short-term

temperature–crime links using distributed-lag and other non-linear models (Heo et al., 2024). This literature provides the main justification for testing whether adding temperature as an exogenous regressor improves our purely crime-based SARIMA forecasts.

Against this backdrop, the present study contributes in two main ways. First, the study offers a modern analysis of monthly crime in Fort Worth using seasonal ARIMA models that explicitly account for strong seasonal patterns as identified in previous work (Islam & Raza, 2020; Delgado et al., 2023). Second, it investigates whether adding temperature as an exogenous variable in an ARIMAX framework improves forecast accuracy, effectively merging traditional time series forecasting with the weather–crime literature (Gamble & Hess, 2012; other recent studies).

3. Data Description

This study uses publicly available crime records from the Fort Worth Police Department Open Data Portal (Fort Worth Police Department, 2025), which provides incident-level data for all reported offenses within the city. The original dataset contains daily crime events from February 2016 through September 2025. To create a consistent time series suitable for monthly forecasting, these daily observations were aggregated into monthly total crime counts, forming a continuous series of 116 monthly observations. The primary variable used in all baseline models is the total number of reported crimes per month.

The table below presents summary statistics for the monthly crime series. Crime levels vary considerably across months, with a mean of 5035.92, a minimum of 3896, a maximum of 5942, and a standard deviation of 345.98. These values already suggest substantial fluctuation in criminal activity over time, which visual exploration helps further clarify.

Statistics	Value
Count (months)	116
Mean	5035.92
Median	5027
Std Dev	345.98
Min	3896
Max	5942
25th Percentile	4816.75
75th Percentile	5265.25
IQR	448.5
Coefficient of Variation	0.07
Skewness	-0.23
Kurtosis	0.64

Table 1. Summary Statistics for Monthly Crime Counts

A plot was generated to examine the structure of the data. The figure below shows the monthly crime levels over the full sample period, revealing clear seasonal fluctuations and occasional sharp deviations.

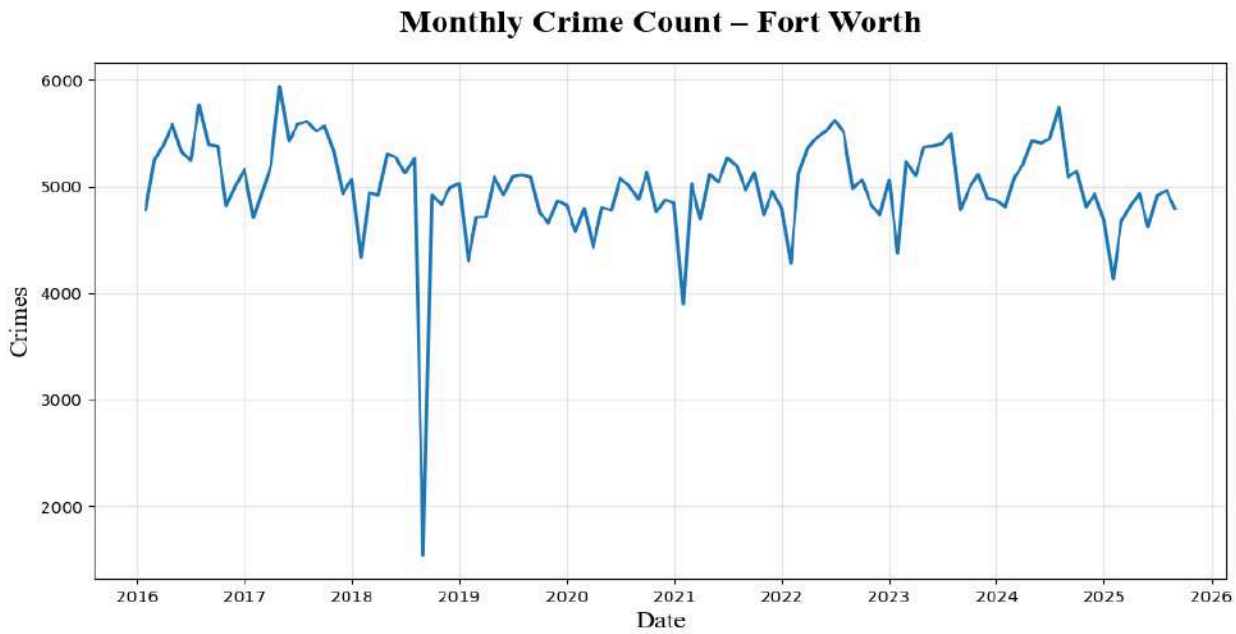

Figure 1: Fort Worth Monthly Crime Count (Before)

One noticeable irregularity was an unusually sharp drop in reported crime for September 2018, which did not align with historical patterns or any documented external event. To avoid allowing this isolated value to distort model estimation, the September 2018 crime count was replaced with a missing value and interpolated using linear interpolation, which estimates the missing point as a straight-line average between the adjacent months.

After interpolation, the series was inspected again to ensure smooth continuity before proceeding with exploratory analysis and model estimation.

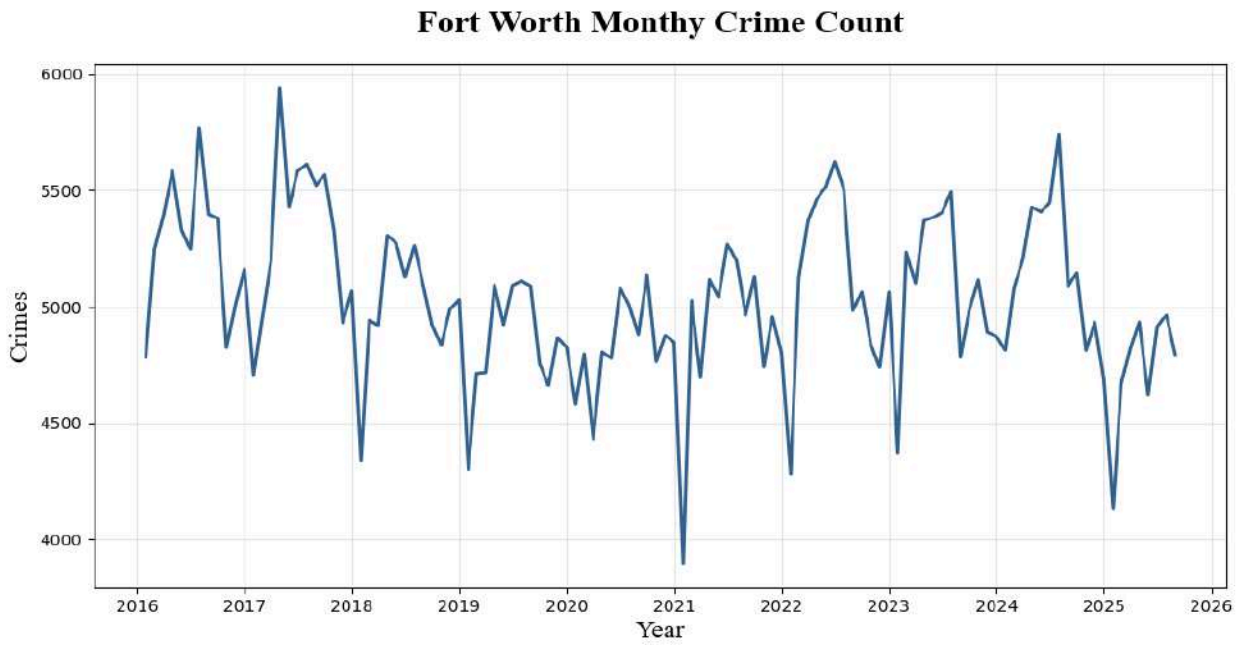

Figure 2: Fort Worth Monthly Crime Count (Interpolated)

Seasonal patterns are further illustrated in the monthly subseries plot below, which shows consistently higher crime levels during the summer and lower activity in winter. These visual patterns suggest strong annual seasonality and possible non-stationarity, which will be formally assessed in the Methodology section.

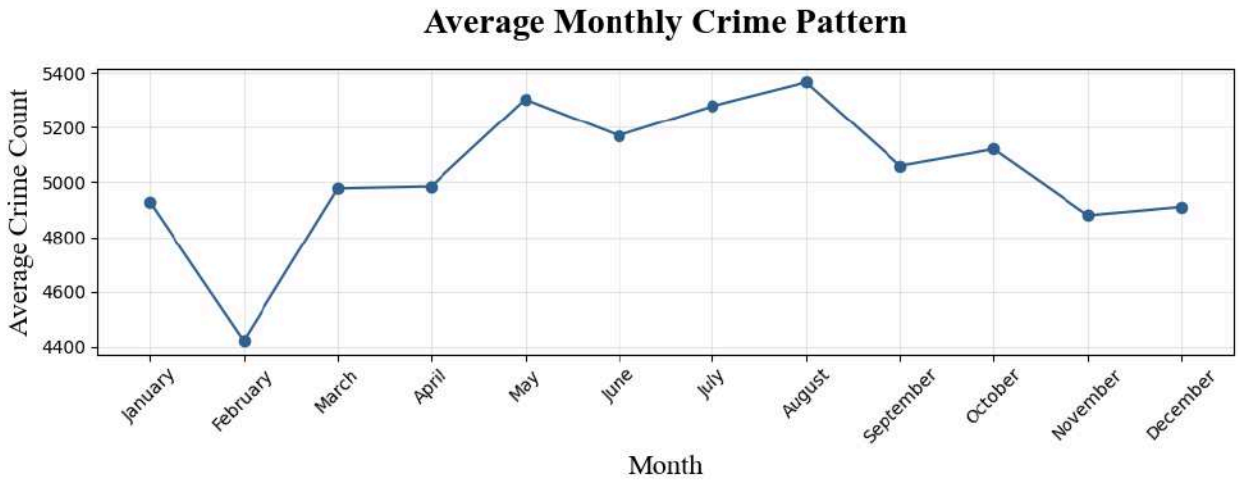

Figure 3: Fort Worth Average Monthly Crime

Overall, the data exhibit meaningful temporal structure, upward or downward trends in certain periods, and seasonality, which motivates the use of seasonal time-series forecasting techniques.

3.1 Temperature with Crime

The analysis focuses solely on this monthly crime count series, as it represents a comprehensive measure of overall criminal activity in the city and aligns with the forecasting objective of predicting aggregate crime levels. In addition to crime data, the monthly average temperature was collected from the National Weather Service Fort Worth Daily Mean Temperature Archive (National Weather Service Fort Worth, 2025). These temperature values were later examined as a potential exogenous regressor to assess whether weather conditions contribute meaningfully to short-term crime dynamics. Although temperature is not included in the core univariate forecasting model, its relationship with crime is explored separately through a dynamic regression model. A combined plot of monthly crime and temperature is shown below.

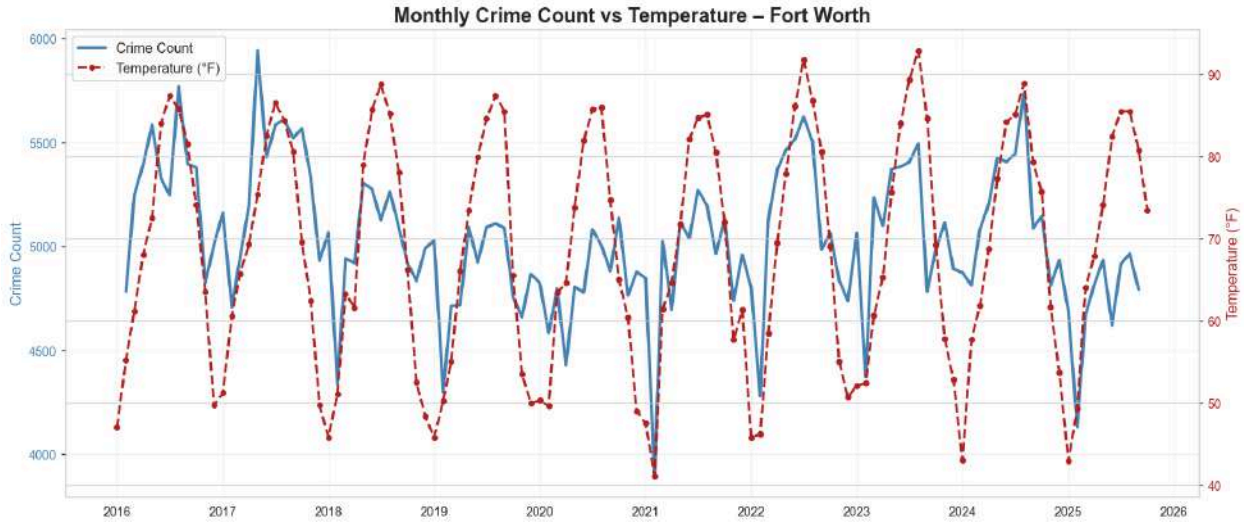

Figure 4: Plotted Temperature and Crime

4. Methodology

After constructing a clean and consistent monthly crime series, the analysis proceeded through several stages: data preparation, exploratory time-series diagnostics to assess trend and seasonality, formal tests of stationarity to determine appropriate levels of differencing, and model identification using patterns in the ACF and PACF. Based on these results, a set of candidate seasonal ARIMA models was estimated and evaluated using information criteria and residual diagnostics. The final selected model was then used to generate out-of-sample forecasts and prediction intervals for monthly crime levels. This section outlines each of these methodological components in detail.

4.1 Data Preparation

During preliminary visualization of the monthly series, an unusual and isolated dip in September 2018 was detected. This value was inconsistent with surrounding months and was not associated with any known event or policy change, suggesting a data-quality issue rather than a genuine decline in crime. To prevent this anomaly from distorting estimates of trend, seasonality, and model parameters, the September 2018 value was replaced with a missing indicator and imputed using linear interpolation between August and October 2018. This approach preserves the continuity of the series without introducing artificial volatility.

The analysis begins with the construction of a consistent monthly time series from the original daily-level crime records. The raw dataset contains one observation per reported crime incident, each tagged with a precise timestamp. For forecasting purposes, the data were aggregated to the monthly level, as monthly frequency provides a clearer view of underlying trends and seasonal patterns while reducing day-to-day noise that is not meaningful for medium-term forecasting.

After cleaning, the monthly crime counts were merged with a corresponding series of monthly average temperatures for Fort Worth, constructed by resampling the daily temperature data to the monthly frequency. This allows for the optional inclusion of temperature as an exogenous regressor in extended SARIMAX models.

The final prepared dataset consists of a complete, monthly, seasonally rich time series from February 2016 through September 2025, suitable for rigorous time-series diagnostics, model identification, and forecasting.

To evaluate forecasting performance, the cleaned monthly crime series was divided into a training set and a test set. The training set included observations from the beginning of the sample through December 2024, providing a sufficiently long history for model estimation and seasonal pattern detection. The remaining observations, covering January 2025 onward, were held out as the test set. This temporal split mirrors a real-world forecasting scenario in which only past data are available when generating predictions, allowing for an unbiased assessment of each model's out-of-sample accuracy.

4.2 Exploratory Analysis

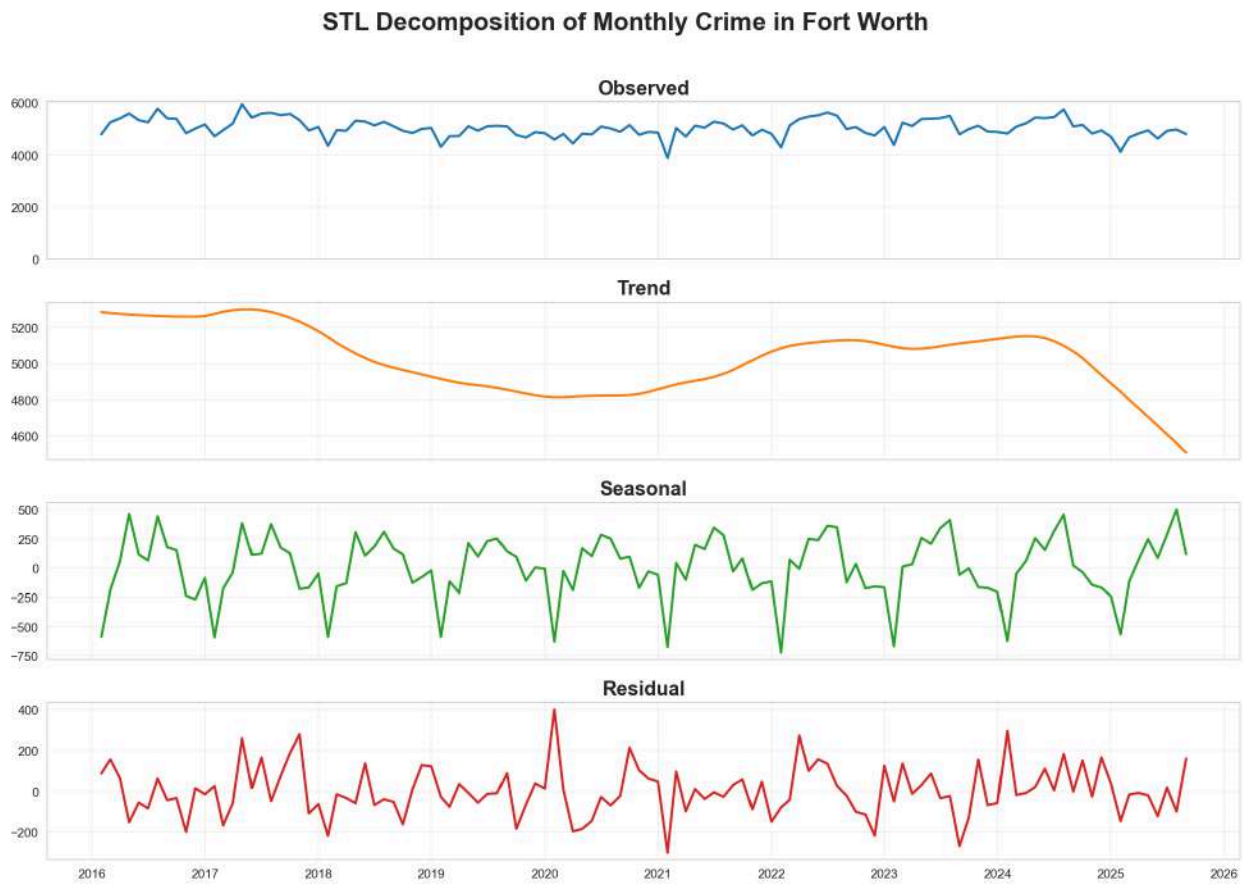

Figure 5: STL Decomposition of Monthly Crime

To better understand the underlying structure of the monthly crime series and guide the selection of an appropriate forecasting model, the data were decomposed using Seasonal–Trend Decomposition via Loess (STL). The STL results, presented in Figure X, reveal three key features of the series. First, the trend component shows a gradual decline in crime from 2016 through 2020, followed by a moderate recovery through 2023 before tapering again in 2025. Second, the seasonal component displays a strong and highly regular annual pattern, with crime consistently peaking during the summer months and falling during winter. The stability of this seasonal cycle across years indicates that a seasonal ARIMA structure with yearly periodicity is

appropriate. Finally, the residual component shows relatively small, short-lived deviations with no persistent autocorrelation, suggesting that much of the systematic variation in the series is captured by the trend and seasonal components.

Visual inspection of the observed series and the residual component from the STL decomposition indicates that the variance of monthly crime counts remains relatively stable over time, and the amplitude of the seasonal cycle does not increase with the level of the series. Because the series exhibits additive seasonality and no evidence of level-dependent variance, no transformation was applied before model identification.

Beyond the trend and seasonal patterns revealed by the STL decomposition, prior empirical research consistently shows that temperature is positively correlated with crime, particularly violent and interpersonal offenses. Warmer conditions are associated with greater outdoor activity and increased opportunities for conflict, leading many studies to document a temperature–crime linkage. To assess whether a similar relationship exists in Fort Worth, a scatterplot of monthly average temperature versus monthly crime count was constructed.

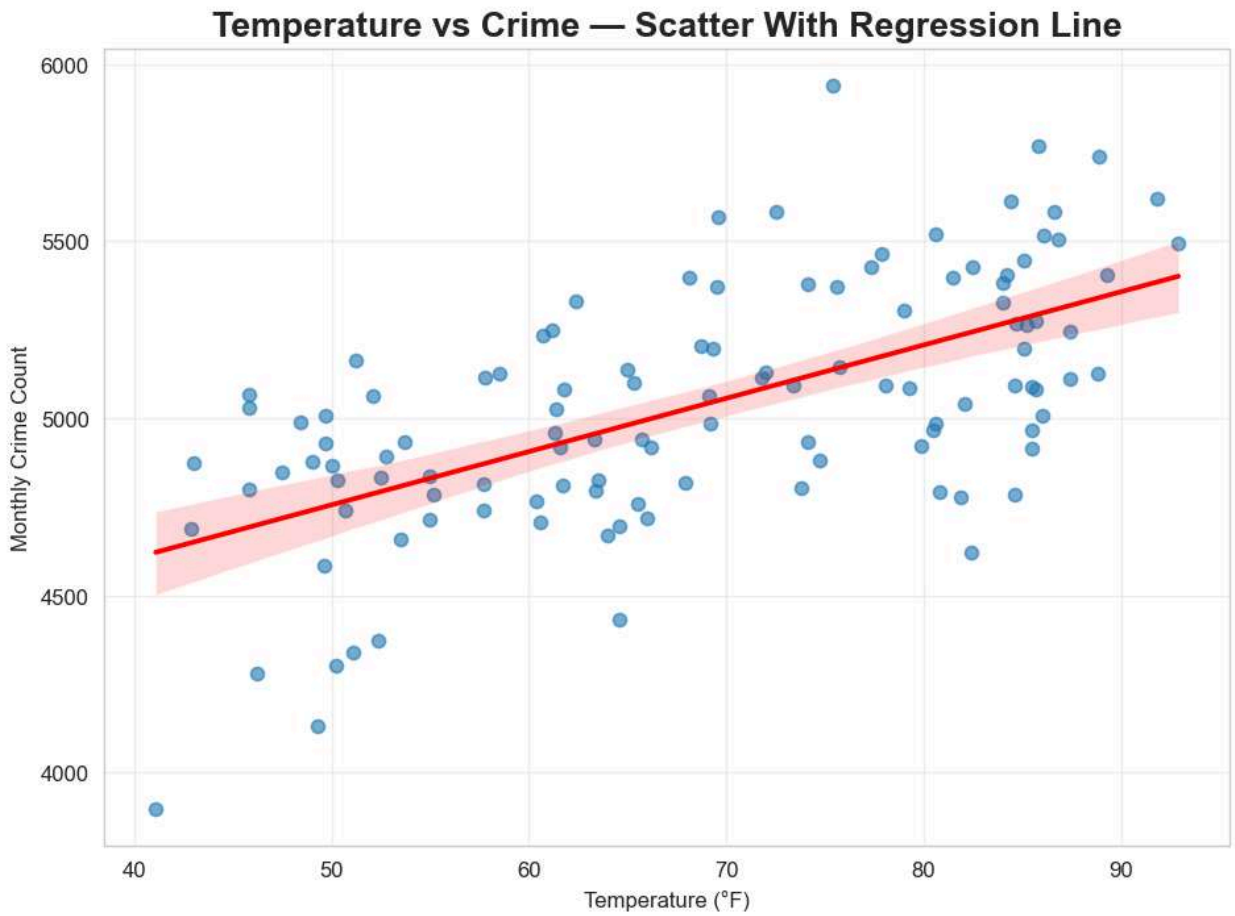

Figure 6: Temperature vs Crime

The upward-sloping regression line indicates a clear positive association, suggesting that months with higher temperatures tend to experience higher crime levels. While this plot alone does not imply causality, it provides preliminary evidence that temperature may carry predictive value, motivating the inclusion of temperature as an exogenous regressor in the ARIMAX model estimated later in the paper.

4.3 Stationarity and Differencing

A key step in specifying an appropriate SARIMA model is determining whether the series is stationary in its mean and variance. Visual inspection of the monthly crime series suggested no strong long-run trend but a clear repeating annual pattern, implying the presence of seasonal

non-stationarity. To formally assess this, the OCSB test implemented through nsdiffs() in R was applied, and the result indicated that one seasonal difference (D = 1) was required to remove the seasonal unit root.

To determine whether additional non-seasonal differencing was required, the augmented Dickey–Fuller (ADF) test was applied to the seasonally differenced series. The test did not reject the null hypothesis of a unit root, suggesting that the series remained non-stationary after seasonal differencing alone. To address this, an additional first difference was applied, resulting in a twice-differenced series.

```
Value of test-statistic is: -3.1384 4.9274

Critical values for test statistics:
      1pct  5pct 10pct
tau2 -3.51 -2.89 -2.58
phi1  6.70  4.71  3.86
```

The ADF test on this transformed series yielded a test statistic of approximately -3.13, which is smaller than the 5% and 10% critical values (–2.89 and –2.55). This indicates rejection of the unit root at conventional significance levels, confirming that the twice-differenced series is stationary.

4.4 Model Identification (ACF & PACF)

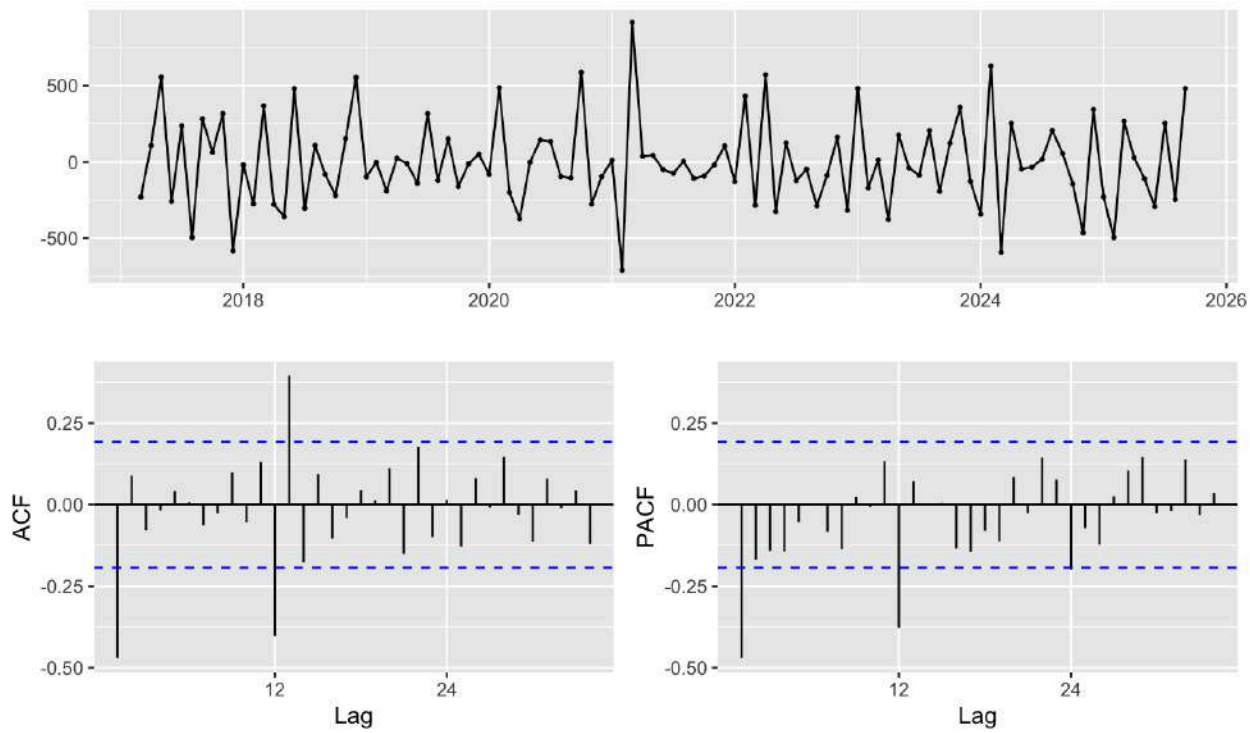

Figure 7: ACF and PACF Plots

Inspection of the ACF and PACF plots of the differenced series reveals strong negative spikes at both lag 1 and lag 12. At the non-seasonal level, both plots exhibit a significant negative autocorrelation at lag 1, suggesting that either an AR(1) term (p = 1) or an MA(1) term (q = 1) or potentially both may be appropriate. At the seasonal level, similar negative spikes at lag 12 in both the ACF and PACF indicate that both a seasonal AR(1) (P = 1) and a seasonal MA(1) (Q = 1) should be considered.

These patterns suggest that several SARIMA structures are plausible, including models with one or both AR and MA components at both seasonal and non-seasonal frequencies. Candidate

models therefore include specifications such as SARIMA(1,1,1)(1,1,1) [12], SARIMA(0,1,1)(0,1,1) [12], SARIMA(1,1,0)(0,1,1)[12], etc.

Because ACF/PACF inspection can be somewhat subjective, an automated model-selection procedure was also employed using auto.arima() from the forecast package. This function evaluates alternative SARIMA specifications under information criteria such as AICc and provides a benchmark against which the manually identified model can be compared. The final model candidates were therefore chosen by combining the insights from the correlation structure with the automated model selection output.

4. Results

All models were estimated using the training sample from February 2016 through December 2024, and their predictive performance was evaluated on the withheld test set consisting of the first nine months of 2025. This structure ensures that the selected model is assessed not only on its in-sample fit but also on its ability to generalize to unseen future data, which is central to the purpose of time-series forecasting.

4.1 Model Estimation

Following the model identification stage, two SARIMA specifications were estimated using the training sample. The first model was obtained through the auto.arima() procedure, which selected an ARIMA(0,1,1)(0,1,1)[12] structure. The second was the manually identified SARIMA(1,1,1)(1,1,1)[12] model, motivated not only by the ACF and PACF patterns of the seasonally differenced training data, but also because it performed better than several alternative SARIMA candidates considered during manual exploration.

The estimated coefficients for both models are presented in the table below.

Model	ar1	ma1	sar1	sma1	σ^2	LogLik	AICc	BIC
ARIMA(0,1,1)(0,1,1)[12]	–	–0.6720 (0.1072)	–	–0.6494 (0.1275)	44,320	–638.82	1283.9	1291.26
ARIMA(1,1,1)(1,1,1)[12]	0.1857 (0.1420)	–0.8085 (0.0908)	0.2731 (0.1218)	–1.0000 (0.2739)	37,348	–636.78	1284.24	1296.27

Table 2: Estimated Coefficients

A comparison of the AICc values shows that the two SARIMA models perform very similarly, with the auto-selected ARIMA(0,1,1)(0,1,1)[12] achieving an AICc of 1283.9 and the manually identified SARIMA(1,1,1)(1,1,1)[12] producing an AICc of 1284.24.

To benchmark the SARIMA specification, an exponential smoothing model (ETS) was estimated on the training dataset. The selected ETS model was ETS(A, N, A), which includes additive seasonality but no trend component.

Model	AIC	AICc	BIC	ME (Train)	RMSE (Train)	MAE (Train)	MPE (Train)	MAPE (Train)	MASE (Train)	ACF1	sigma
ETS(A,N,A)	1641.708	1646.982	1681.8	-5.8427	180.3554	142.1723	-0.2137	2.8488	0.6883	0.09815	193.4548

Table 3: ETS Model Summary Statistics

Although ETS(A, N, A) provides a reasonable seasonal decomposition, its substantially higher AICc (1646.98), non-white residuals, and weaker accuracy metrics indicate inferior performance relative to the SARIMA models. Therefore, ETS was not considered for final forecasting.

After estimating both models, diagnostic checks were conducted to verify whether the residuals behaved like white noise, as required for a well-specified ARIMA model. Figures below display the standardized residuals, ACF of residuals, and the residual histograms for the auto.arima model and the manually selected SARIMA(1,1,1)(1,1,1)[12] model.

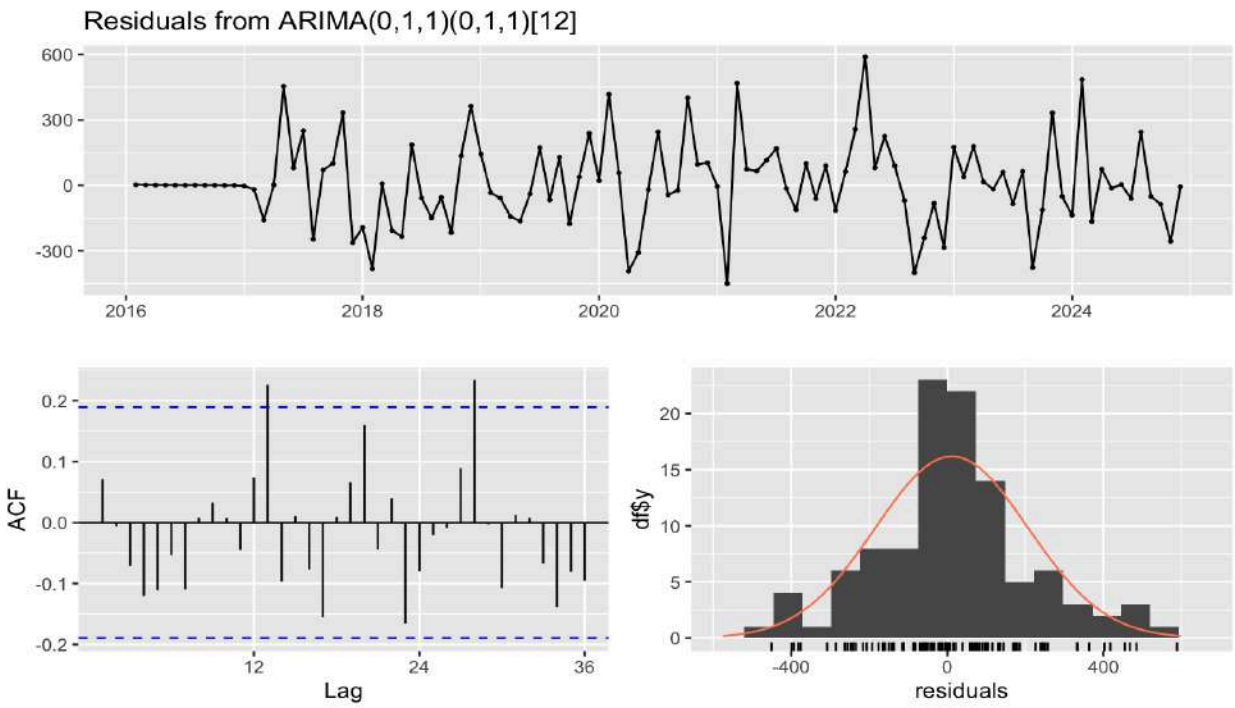

Figure 8: Residuals ARIMA(0,1,1)(0,1,1)[12]

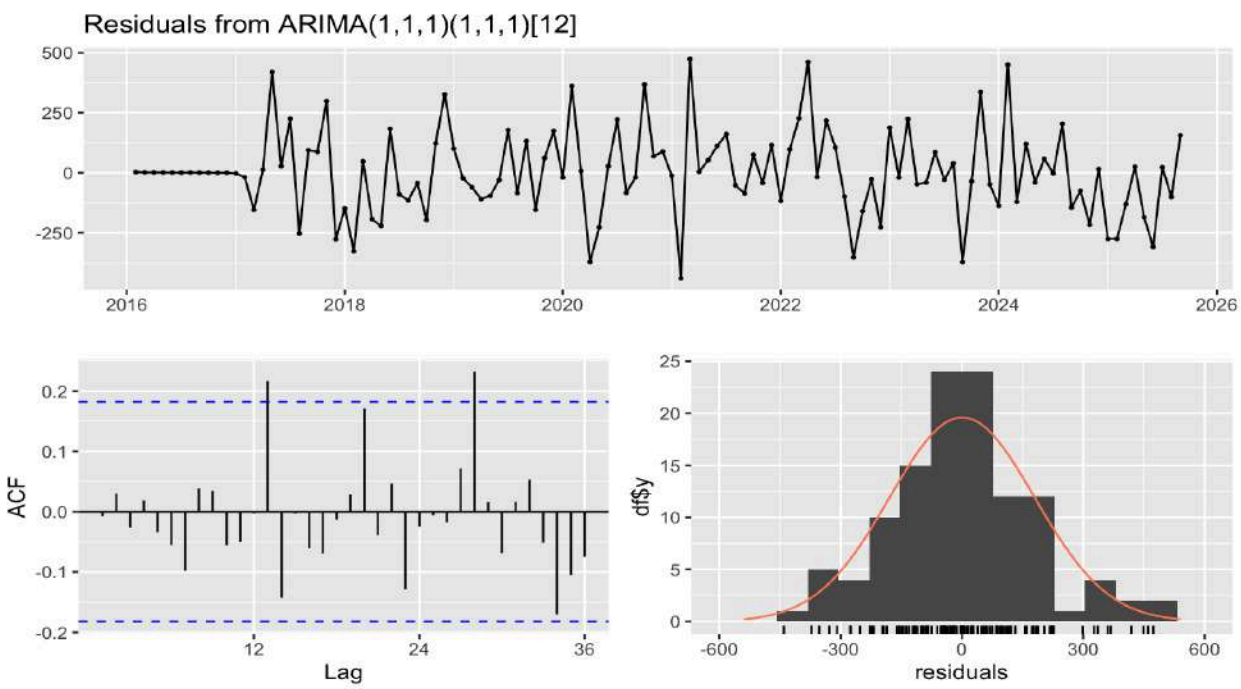

Figure 9: Residuals ARIMA(1,1,1)(1,1,1)[12]

For both models, the residuals are centered around zero with no obvious patterns over time. The ACF plots show no significant spikes beyond the 95% bounds, indicating that neither model leaves strong autocorrelation unexplained.

Model	Ljung–Box Q*	df	p-value
SARIMA(0,1,1)(0,1,1) [12]	22.668	19	0.2523
SARIMA(1,1,1)(1,1,1) [12]	20.417	19	0.3699

Table 4: Ljung-Box Test Summary

The table above reports the Ljung–Box Q-statistics and associated p-values for both models. In each case, the p-values comfortably exceed the 5% significance threshold, meaning the null hypothesis of no residual autocorrelation cannot be rejected. This confirms that the residuals behave as white noise and that the fitted models are statistically adequate.

However, the manually chosen SARIMA model produces slightly better diagnostics: a higher Ljung–Box p-value and residuals that more closely align with the normal distribution. These results, along with its superior AICc, make it the preferred model for forecasting.

Note: The first 13 residual values appear as zero due to the combined non-seasonal and seasonal differencing (d = 1, D = 1, period = 12). Residuals cannot be computed for observations removed through differencing, so these zeros are placeholders and not interpreted in diagnostics.

4.2 Model Validation

To evaluate forecasting performance, both models were assessed using a standard training–testing split, where the last nine months of the dataset (January–September 2025) were

withheld for out-of-sample validation. Accuracy metrics for both the training and testing periods are reported in the table below.

Model	Dataset	RMSE	MAE	MAPE	MASE	ME	ACF_1	Theil's U
ARIMA(0,1,1)(0,1,1)[12]	Training	195.21	140.61	2.82%	1	11.34	0.071	–
	Test	484.34	460.7	9.76%	2	–460.70	0.0715	1.4707
ARIMA(1,1,1)(1,1,1)[12]	Training	177.24	128.55	2.57%	0.622	14.07	–0.0233	–
	Test	467.59	449.97	9.55%	2.18	–449.97	–0.0025	1.4103

Table 5: Train and Test Accuracies

In terms of in-sample performance, the SARIMA(1,1,1)(1,1,1)[12] model outperforms the auto.arima model. Its RMSE, MAE, and MAPE values are smaller, which further confirms its ability to capture the underlying dynamics of monthly crime counts.

When evaluated on the test set, both models deteriorate in accuracy, which is expected because the forecasted period contains a noticeable decline in early 2025 that neither model predicts well. Nevertheless, the manually specified SARIMA(1,1,1)(1,1,1) model again performs better, achieving lower RMSE, MAE, and MAPE values and a lower Theil's U statistic. The auto.arima model, by comparison, exhibits higher forecast variance and larger magnitude forecast errors.

4.3 Model Selection

Although the AICc values of the two models are extremely close, the SARIMA(1,1,1)(1,1,1)[12] model achieves a lower in-sample error and a substantially reduced innovation variance, indicating a tighter fit to the underlying crime dynamics. More importantly, the model performs consistently better in the out-of-sample evaluation period, producing lower RMSE, MAE, MAPE, and Theil's U statistics. Because out-of-sample forecasting accuracy is the most

meaningful criterion for model comparison, the SARIMA(1,1,1)(1,1,1) [12] specification is selected as the preferred model for generating the final forecasts.

4.4 Forecasting

Using the final selected model, SARIMA(1,1,1)(1,1,1) [12], an 18-month forecast of monthly crime counts was generated. The figure below shows the training data (black), the test period (blue), and the model's forecast (red) along with 95% prediction intervals (green and orange dashed lines).

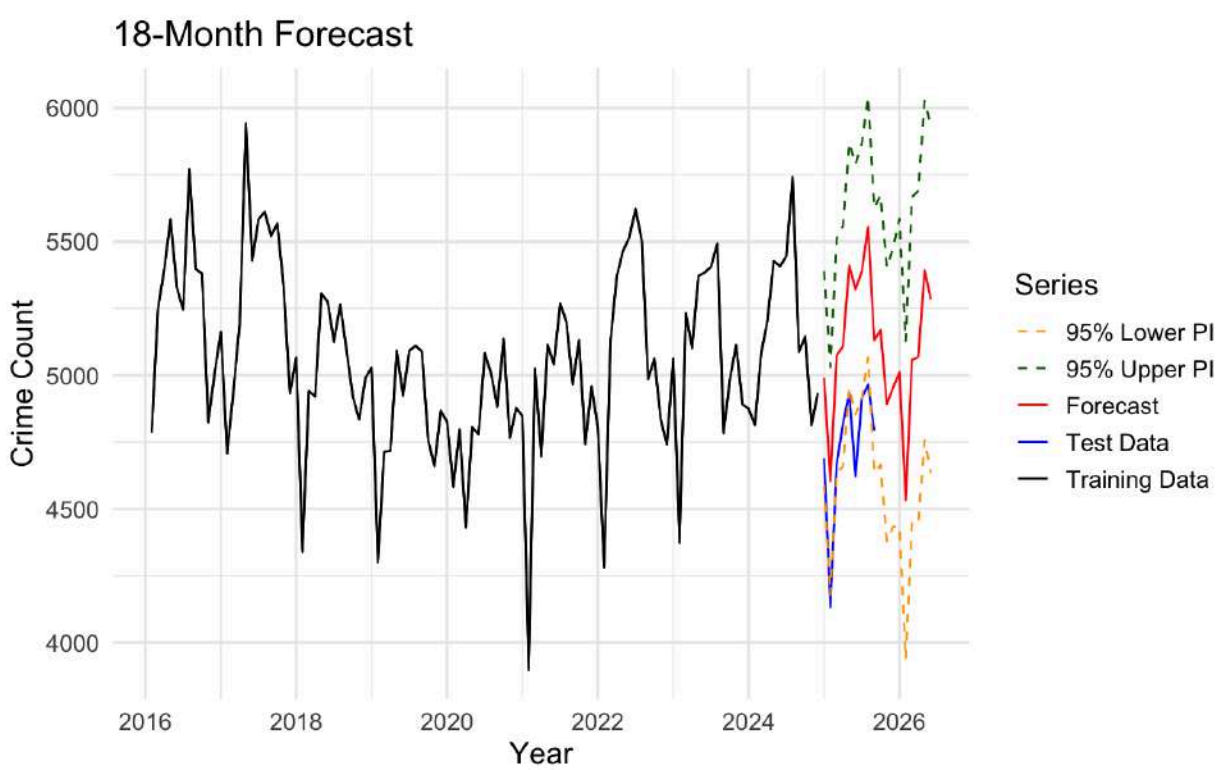

Figure 10: Forecast Using ARIMA(1,1,1)(1,1,1)[12]

The forecast extends the seasonal pattern observed in the historical data: crime is expected to rise during summer months and fall during winter, with most predicted values remaining between roughly 4,800 and 5,500 crimes per month. The 95% prediction intervals widen gradually over

time, reflecting increasing uncertainty, but they remain centered around a stable seasonal mean, suggesting no major structural change.

Importantly, the test data fall well within the forecast distribution, indicating that the chosen model generalizes reasonably and captures the key seasonal dynamics. Overall, the 18-month projection provides a realistic continuation of recent crime patterns in Fort Worth and supports the model's suitability for short-term forecasting.

5. Incorporating Explanatory Variables

Although the SARIMA-based forecast provides a reasonable projection of future crime patterns, it is important to acknowledge that crime rates are influenced by a wide range of social, economic, and environmental factors. Research consistently shows that variables such as unemployment, income inequality, policing intensity, economic distress, and even weather conditions shape crime dynamics at both national and local levels.

However, obtaining reliable monthly data for these factors, especially at the city level for Fort Worth, is difficult. Many socio-economic indicators are reported annually, with limited granularity, while others are not publicly available or lack consistent historical coverage.

Temperature, however, is an important exception. Numerous studies document a strong relationship between heat and crime, particularly violence, as higher temperatures tend to increase outdoor activity and aggressive behavior. Monthly temperature data are readily available for the Dallas–Fort Worth region, making it a practical and meaningful candidate for improving the baseline SARIMA forecast.

For this reason, the next section extends the analysis by incorporating average monthly temperature as an exogenous regressor. This allows us to evaluate whether adding weather information improves forecast accuracy and provides additional insight into the drivers of crime levels in Fort Worth.

5.1 ARIMAX Model with Temperature

To test whether temperature improves forecasting performance, an ARIMAX model was fitted using average monthly temperature as an external regressor. The selected specification was ARIMAX(1,1,1)(1,0,1) [12]. The table below reports the parameter estimates.

Parameter	Estimate	Std. Error
ar1	0.2568	0.1353
ma1	-0.8623	0.0735
sar1	0.8802	0.1025
sma1	-0.4921	0.2124
Temperature (xreg)	16.4609	3.5206

Table 6: ARIMAX Estimated Parameters

The temperature coefficient is positive and statistically significant, suggesting that a 1°F increase in average monthly temperature is associated with roughly 16 additional reported crimes, holding other dynamics constant.

```
        Ljung-Box test

data:  Residuals from Regression with ARIMA(1,1,1)(1,0,1)[12] errors
Q* = 19.285, df = 17, p-value = 0.3124

Model df: 4.   Total lags used: 21
```

Residuals were well-behaved. The Ljung–Box statistic (Q*=19.285, p=0.3124) indicates no remaining autocorrelation. Thus, the ARIMAX model is statistically significant.

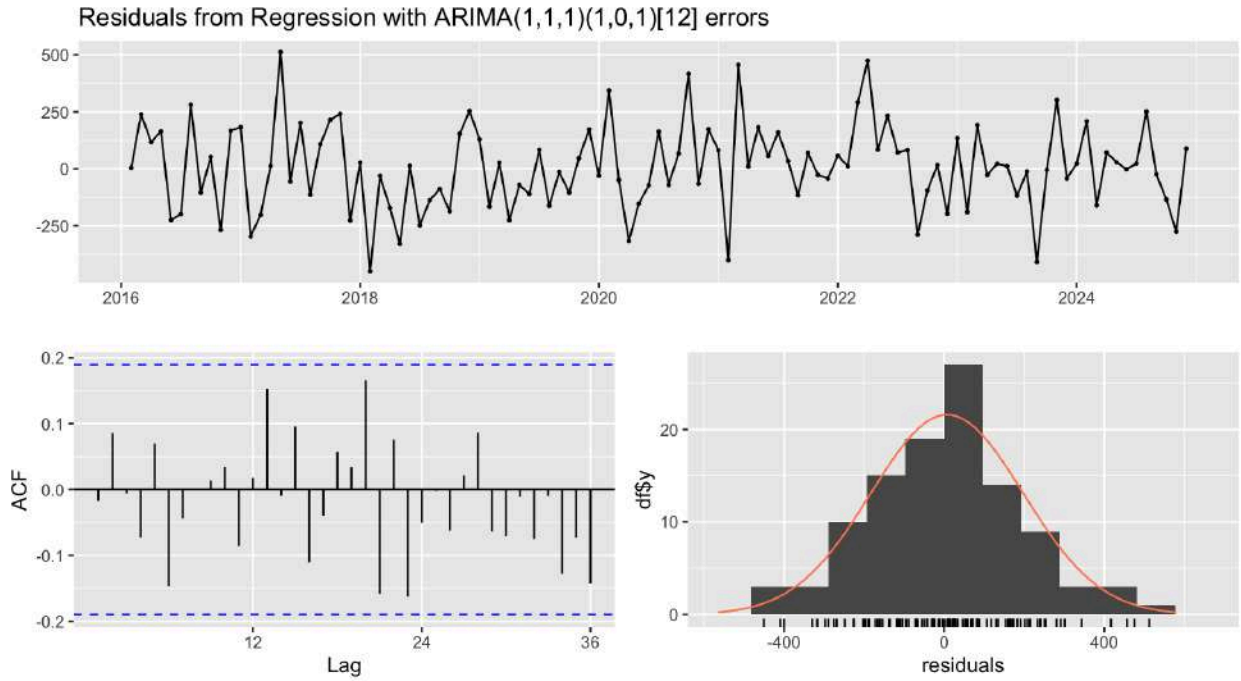

Figure 11: Residuals ARIMA(1,1,1)(1,0,1)[12]

The residual time-series plot shows no systematic patterns or visible autocorrelation, suggesting that the model has captured the main temporal dynamics in the data. The ACF of residuals displays no significant spikes beyond the 95% bounds, indicating the absence of remaining serial correlation.

The distribution of residuals is approximately symmetric and unimodal, aligning reasonably well with the normal density curve.

Dataset	RMSE	MAE	MAPE	Theil's U
Training	189.76	147.75	2.95%	—
Test	489.9	464.07	9.86%	1.512

Table 7: ARIMAX Train and Test Errors

Although temperature is significant, the ARIMAX model does not outperform the SARIMA(1,1,1)(1,1,1) [12] model. Test RMSE and MAPE are slightly higher for ARIMAX, indicating weaker out-of-sample accuracy. This suggests that most of the predictability in crime is already captured by strong seasonal patterns, and temperature alone does not substantially improve short-term forecasting performance.

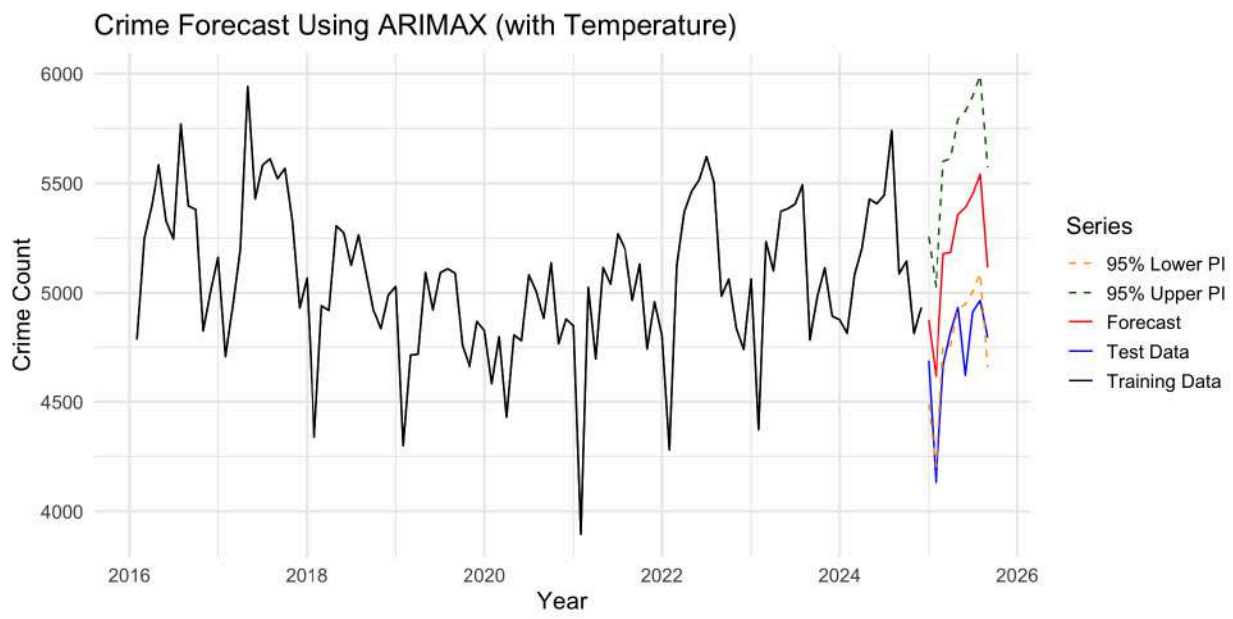

Figure 12: Forecast Using ARIMAX(1,1,1)(1,1,1)[12]

The forecast plot (attached) shows the fitted ARIMAX forecast alongside the training data, test data, and 95% prediction intervals. The model successfully tracks the rebound in crime visible in early 2025, with forecasted values closely following the observed test data during seasonal peaks and troughs. However, the prediction intervals widen as the horizon increases, indicating greater uncertainty in long-term projections.

Overall, the model provides a reasonable forward-looking outlook, indicating a seasonal rise in crime during warmer months followed by a tapering in later periods, but also highlights the limits of forecasting accuracy when using a single exogenous regressor such as temperature.

6. Conclusion

This study analyzed monthly crime patterns in Fort Worth using time-series methods, beginning with SARIMA models that captured the strong seasonal structure evident in the data. After conducting model identification, estimation, and validation using a training–test split, the SARIMA(1,1,1)(1,1,1) [12] specification produced the most reliable forecasts, with well-behaved residuals and comparatively low test error. Because prior research suggests that weather influences crime, an extended ARIMAX model incorporating temperature was also estimated. While temperature was statistically significant and positively associated with crime, its inclusion did not meaningfully improve forecast accuracy relative to the SARIMA benchmark. A comparison with exponential smoothing (ETS) models further confirmed that SARIMA remained the best-performing framework for this dataset.

There are a few aspects that could have strengthened the analysis. Working with daily data rather than monthly totals may have revealed more short-term movements in crime, where popular models such as Facebook's Prophet could have been tested, as it is designed to handle high-frequency, seasonal patterns more flexibly. It also would have been helpful to explore nonlinear or machine-learning approaches, since crime dynamics may not always follow the linear structure assumed by traditional time series models.

These results also raise several questions for future research. For example, would different crime categories respond differently to temperature or to other external factors not included here?

Would forecasts improve if additional variables, such as unemployment, policing activity, or school schedules, were included?

Overall, the findings show that crime in Fort Worth follows a predictable seasonal pattern and that traditional seasonal ARIMA models provide a strong basis for short-term forecasting. Although temperature contributes to explaining variation, its predictive value, at least in monthly aggregates, appears limited compared to the underlying seasonal dynamics.

7. Limitations

Several limitations of the study should be acknowledged. First, the analysis relies solely on police-reported crime counts, which may be affected by reporting practices, classification changes, or administrative delays. Second, temperature is the only exogenous variable incorporated, even though crime is known to be influenced by a wide range of socio-economic and environmental factors, such as unemployment, income inequality, policing intensity, school schedules, and major events, that were not included due to data availability constraints. Third, model selection was based on traditional linear time-series approaches; although effective, these models may not fully capture evolving crime dynamics or complex non-linear relationships that machine-learning methods can exploit. Finally, the study evaluates forecasts over a relatively short out-of-sample period, which limits conclusions about long-term predictive performance.

8. Use of AI

ChatGPT (OpenAI, 2025) was used for grammar correction, language refinement, and coding guidance

9. References

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